P0161 OXYGEN-SENSOR HEATER GUIDE

P0161 — Bank 2 Sensor 2 heater circuit

CONFIRMED The OBD-II definition is a heater-circuit fault on Bank 2, Sensor 2. On the California 4.0L layout that position is the **rear downstream sensor**: after the rear mini-catalyst and before the main catalyst. **LIKELY** location. Count the sensors on this truck before you order a part. A two-sensor federal truck does not use this layout.

What the heater does

Each sensor has four circuits: heater power, heater control, signal return, and the sensor signal. The heater brings the zirconia element up to operating temperature. Until it is hot, the PCM cannot trust the signal. P0161 means the heater circuit failed the monitor. It does not grade the air-fuel signal.

What not to do first

Do not buy a sensor because the code is stored. Do not swap it with another sensor. Do not clear the code to "see if it comes back" before the tests below are written down. Do not unplug a sensor on a hot pipe. Exhaust iron stays burn-hot long after shutdown.

Work in this order

1. **Cold visual.** Engine cool. Find the rear downstream sensor by tracing the rear downpipe, not by guessing at the connector colors. Look at the connector shell, the latch, the terminals, and the first 12 inches of wire.
2. **Fuse and ASD feed.** Identify the ASD relay and its fuse from the under-hood power-distribution-center lid. Do not use a fuse number from memory.
3. **Heater resistance** on the sensor side, sensor at room temperature. Screening band about 4–10 Ω.
4. **Heater power,** about 10–12 V, measured while the feed is actually energized.
5. **PCM control** on the other heater wire. Repair the failed side. Replace the sensor only when the element itself is open or shorted, or the connector on the sensor pigtail is destroyed.

Oil changes the order

If oil is inside this connector or coating this sensor, the leak is a fire issue and a parts issue. Guide 06. Clean and repair the leak before a new sensor is installed. An oil-soaked connector can cause P0161 even when the heater element is intact. **POSSIBLE** **TEST REQUIRED**

Wire colors are not printed in this guide. Published colors for earlier 4.0L years do not reliably match every 2004 connector. Identify the heater pair by the resistance test, then mark the cavities. A factory diagram for this VIN is the authority if the connector is damaged or the pins are not obvious.

P0161 · CONNECTOR, FUSE, HEATER RESISTANCE

Connector, fuse, and resistance

Connector and wiring inspection

- Latch broken, or the plug seated but not locked.
- Green corrosion, oil, or water in the shell.
- Terminals backed out or spread open. A gentle tug on each wire.
- Insulation melted or brittle where the harness passed the pipe.
- Harness clips missing, so the wire rests on the manifold or the downpipe.

Pull on the connector body, never on the wire entering the sensor. Do not grease the plug. Do not solder the sensor leads. A twisted, taped splice next to the exhaust will fail again. If a wire is broken, repair it with a sealed splice and reroute it in the original clips, with slack for engine rock.

Fuse and ASD feed TEST REQUIRED

Heater power on this Chrysler family is fed from the automatic-shutdown (ASD) relay circuit, not from a constant key-on wire. Open the under-hood center and read the lid. Find the ASD relay and the fuse that feeds it.

1. Remove that fuse and look at the element. Also measure continuity across the fuse. A fuse can look intact and be open.
2. With the fuse installed, key on, check for voltage on both fuse legs during the moment the circuit is powered. One live leg and one dead leg means the fuse is open.
3. If every oxygen-sensor heater feed is dead, suspect the fuse, the ASD relay, or the ASD output. If the other sensors have power and this one does not, suspect the branch wire to Bank 2 Sensor 2.

Heater resistance — sensor cold TEST REQUIRED

Screening band for this job: about 4–10 Ω at room temperature, measured on the sensor connector, not through the engine harness. Earlier 4.0L factory tests call for roughly 4–7 Ω, and a 1997 Grand Cherokee test calls for about 4.5 Ω to 7 Ω. Those limits are LIKELY cousins of this sensor, not a 2004 specification sheet in hand. Resistance rises as the sensor gets hot. A hot reading outside the band does not condemn the part. Let it cool.

Reading, sensor side, cold	Meaning	Next move
About 4–10 Ω on one pair only	Heater element is electrically plausible. IN RANGE	Do not replace yet. Test power and control.
OL, or many thousands of ohms, on every pair	Open heater, or you are not on the sensor connector.	Recheck that you are on the sensor pigtail. If still open, the sensor heater has failed.
Near 0 Ω	Shorted heater element.	Sensor is bad. Check the fuse too, because a short can open it.
Two different pairs both read a few ohms	You may be bridging signal and heater, or the element is shorted internally.	Stop and sort the cavities before you condemn the PCM.

How to find the heater pair without a diagram: the heater is the only pair that reads a handful of ohms when cold. The signal terminals of a cold narrowband sensor do not. Write down which two cavities those are. The engine-harness side is where voltage is tested, and it must mate to those same two cavities.

P0161 · POWER AND PCM CONTROL

Heater power and PCM control

Key-on is not always "circuit powered"
A steady key-on, engine-off reading of 0 V can be a false failure. On this Chrysler family the ASD relay often pulls in for about a second at key-on, then drops out if the engine is not turning. Earlier 4.0L procedures do specify about **10–12 V, key on, engine off**, on the heater feed. Use that number. Take the reading while the relay is actually on: during that first second, while cranking, or with the engine running. Backprobe. Do not pierce the wire.

Power test TEST REQUIRED

1. Sensor unplugged or backprobed at the harness. Meter negative on a clean engine ground, or on the battery negative.
2. Probe the harness cavity that mates to one heater terminal. Key on. Watch the meter. Repeat on the other heater cavity.
3. The feed side should show about 10–12 V while the ASD is energized. The control side may show voltage only when the PCM is not pulling it down, or a pulsing value. Do not expect both wires to sit at a hard 12 V.
4. Compare with a heater feed at another sensor on the same truck. Power at the others and none here isolates the branch to Bank 2 Sensor 2.

Control / ground test TEST REQUIRED

The PCM switches the heater ground. It is often a pulse, not a solid ground, so a meter may show a changing voltage instead of 0.00 V. With the engine running and the heater feed proven:

- A control wire that the PCM never pulls, while another sensor's heater control does show activity, points at this wire or at the PCM driver for this sensor.
- A control wire that is shorted to ground with the key off and the PCM connector considered can keep the heater on or blow the fuse. Check for a rub-through to the body or the block.
- A control wire shorted to battery voltage will set a circuit fault and can damage a driver. Do not connect it to ground to "make the code go away."

Do not leave a jumper from the control wire to ground. A heater of about 5 Ω draws roughly 2.5 A at 12 V. These heaters are pulsed. A sustained hard ground can overheat the element and the wire. Do not use a jumper as a repair.

Heater ohms	10–12 V feed	PCM control	Action
Open or shorted	Present on the harness	Not the cause	Replace the sensor after oil and connector problems are corrected.
In the 4–10 Ω screen	Missing at this sensor only	—	Repair the feed wire or its connection. Recheck the fuse.
In range	Missing at all sensors	—	Fuse, ASD relay, or ASD output. TEST REQUIRED
In range	Present	Never switches, others do	Repair the control wire. Condemn the PCM only after the wire and the connector test good.
In range	Present	Switches	Look for an intermittent: oil, a spread pin, a wire that opens when the engine rocks. Do not replace the sensor on the code alone.

P0161 · PARTS AND INSTALLATION

When the sensor is actually the part

Replace Bank 2 Sensor 2 when the cold heater is open or shorted, or the sensor-side connector is damaged beyond a terminal repair, and the harness feed has been proven. Fix an oil leak first if oil is reaching this sensor. Guide 09 has the part-number warnings. Short version:

NOT CONFIRMED **NTK 23142** and **Mopar 56029001AB** were checked against published fitment and did not come back as confirmed parts for a 2004 Grand Cherokee 4.0L Bank 2 Sensor 2. Do not buy either number from this sheet. Give a dealer the VIN and the position: rear pipe, after the rear mini-cat. NTK is the usual original supplier of Mopar sensors of this era. **LIKELY** from parts-trade reporting, not a Chrysler letter. This PCM has a poor field reputation with no-name and many Bosch sensors. **POSSIBLE** pattern. Prefer a new NTK or Mopar sensor for the exact bung.

Installation

1. Exhaust fully cool. Raise and support the truck if the bung is reached from below. Wheels chocked, stands rated for the vehicle.
2. Unplug the pigtail from the engine harness. Free the clips. Do not twist the sensor by the wire.
3. Remove it with an oxygen-sensor socket. If it is seized, use penetrating oil on the threads only, not in the connector, and shock the hex. Heat is a last resort on a cold, degreased pipe, with a fire extinguisher at hand and no oil film nearby.
4. A new sensor is already coated. Do not add anti-seize. Extra compound changes the torque and can contaminate the element. Published install torque in the WJ sensor guide is **22 ft-lb (30 N·m)**. **LIKELY** Use a torque wrench. Do not reef it with a breaker bar.
5. Clip the harness back where it was. The wire must not touch the pipe. Connect the plug until the latch clicks.
6. Do not clear codes until Guide 07 says so. After the rest of the repairs, start, confirm there is no fresh oil on the sensor, and rescan. A heater code that returns on the first start is still a circuit problem, not a part that “needs to set.”

Leave the sensor alone when

Ohms are in the screening band, 10–12 V is present while the ASD is on, and the PCM control switches. Spend the time on the connector tension and the harness flex instead. Replacing a good downstream sensor will not fix a high idle, a vacuum leak, or a glowing manifold.

Downstream sensors on this package also help the PCM watch the mini-cats. That function is not what P0161 tests. Signal codes (for example P0157, P0158, P0160) are a different job. Do not retune or delete a sensor to put the light out. This is a California-emissions engine. Repair the circuit.